

Linear Regression

EES 4891/5891

Probability & Statistics for Geosciences

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Getting Started

Getting Started

- Accept the GitHub Classroom assignment at <https://classroom.github.com/a/V0XCBU-0>



Learning Goals

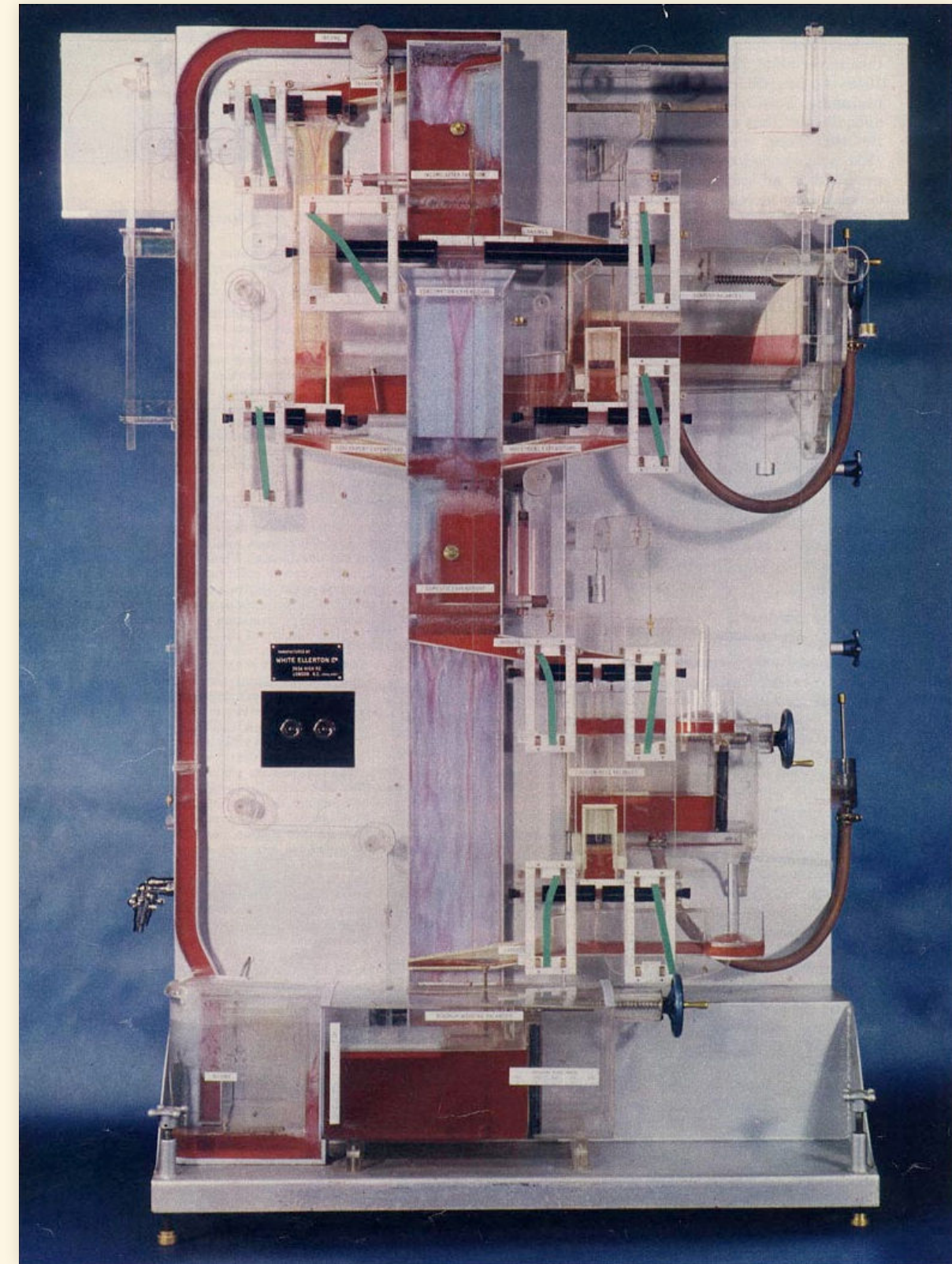
Learning Goals

- Understand what a statistical model is
- Understand what a linear-regression model is
 - Regression versus least-squares
- Be able to use R to
 - Fit a linear-regression model
 - Use a linear regression model to make predictions
- Know how to test the quality of a model by analyzing residuals
- Know how to test the quality of a model with Analysis of Variance (ANOVA)

Statistical Modeling

What is a Model?

- What is a model?
 - There are many kinds of models
 - Mathematical models
 - Computer models
 - System-dynamics models
 - Discrete-event models
 - Stochastic-dynamic models
 - Statistical models
 - Physical models
 - ...



MONIAC: A physical model of the British national economy
(Photo: Wm. Vandivere, *Fortune*, March 1952, p. 100)

What is a Model?

- A model is a purposeful, simplified representation of reality
- Modeling is something we do all the time because we never have enough data, time, and resources to represent every detail.
- “Without models, there are no data.”

Without models, there are no [climate] data. ... I mean this literally. Today, no collection of signals or observations ... becomes global in time and space without first passing through a series of data models. ... Everything we know about the world's climate—past, present, and future—we know through models.

— Paul N. Edwards, A Vast Machine (MIT Press, 2010).

Statistical Models

- What is a statistical model?
 - A model that represents:
 - Distributions of stochastic variables (e.g., measurements)
 - Relations among variables (e.g., covariance)
 - We generally expect a statistical model to be useful for making predictions, based on past observations

- Mathematical formulation

$$X \sim \mathcal{D}(\Theta)$$

X represents observations, and the model says they are drawn from a statistical distribution $\mathcal{D}(\Theta)$, with parameters Θ .

- We can't predict the specific values of a sample X , but we expect that
 - As the # of observations becomes large,
 - The distribution of X will increasingly look like the parent distribution $\mathcal{D}$.
 - This is the Law of Large Numbers

Global Warming from Satellite Measurements

1. Satellites observe intensity of microwave radiation from earth
 - Model of bolometer instrument to convert voltage to microwave intensity
2. Microwave intensity is an integral of emissions from all layers of atmosphere
 - Mathematical model of the atmosphere to calculate emissions from each layer
3. Mathematical physics model: Stefan-Boltzmann Law: $I = \sigma_{\text{SB}} \epsilon T^4$, where
 - I is the emitted microwave intensity,
 - T is the temperature in Kelvin,
 - ϵ is the microwave emissivity,
 - σ_{SB} is the Stefan-Boltzmann constant
 - Calculate T from I .
4. Assemble snapshots of microwave intensity for each spot over the earth into a map of the whole earth.
 - Some data will be missing
5. Statistical model of global average temperature from available data
6. Statistical model of monthly data, over many years, to assess warming trend.
 - Account for random variation
 - Account for autocorrelation (warm years tend to follow warm years)

Linear Regression

Linear Regression

- What is Regression?

- Statistical modeling

- Observations:

- N observations:

- *Dependent* variable $Y = \{y_i\}$,
 $i = 1, \dots, N$

- Independent variables (predictors)
 $X = \{x_{i,j}\}, j = 1, \dots, p$

- Statistical Model:

$$Y = f(X) + \varepsilon$$

$$\varepsilon \sim \mathcal{N}(0, \sigma)$$

- ε represents randomness (residuals, noise, errors, etc.)

- Predictions:

- New observations of X

- Predict what Y should be for the new X :

$$E(Y|X) = f(X)$$

- Special case: Linear Regression

$$E(y_i|x_{i,j}) = f(x_{i,j}) = \beta_0 + \sum_{j=1}^p \beta_j x_{i,j}$$

- Linear Regression vs. Least-Squares:

- Same math, different emphasis

- Least-squares: estimate relationships in observed data

- Regression: predict future data.

Linear Regression in R

- Paleoclimate data: from T.J. Porter *et al.*, *Quaternary Res.* **80**, 167 (2013), DOI: [10.1016/j.yqres.2013.05.004](https://doi.org/10.1016/j.yqres.2013.05.004).

- Load the data:

```
paleo <- read_rds("tree-rings-2.rds")
glimpse(paleo)
```

```
## Rows: 763
## Columns: 3
## $ year_ad      <dbl> 1245, 1246, 1247,...
## $ t_obs        <dbl> NA, NA, NA, NA, N...
## $ growth_index <dbl> 1.41, 1.00, 1.17,...
```

- Tree ring measurements for years 1245 to 2007
- Temperature measurements for years 1893 to 2007.
 - Use these to estimate a linear statistical model
 - Then predict temperature for the other years from measured tree-ring growth index

- Estimate Model

```
paleo_model <- lm(t_obs ~ growth_index, data = paleo,
                  na.action = na.exclude)
summary(paleo_model)
```

```
##
## Call:
## lm(formula = t_obs ~ growth_index, data = paleo, na.action =
na.exclude)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.52464 -0.62025 -0.03916  0.66740  1.65240
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    3.4417     0.3516   9.789  < 2e-16 ***
## growth_index    2.0820     0.2706   7.694 8.52e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8676 on 104 degrees of freedom
## (657 observations deleted due to missingness)
## Multiple R-squared:  0.3627, Adjusted R-squared:  0.3566
## F-statistic: 59.2 on 1 and 104 DF, p-value: 8.517e-12
```

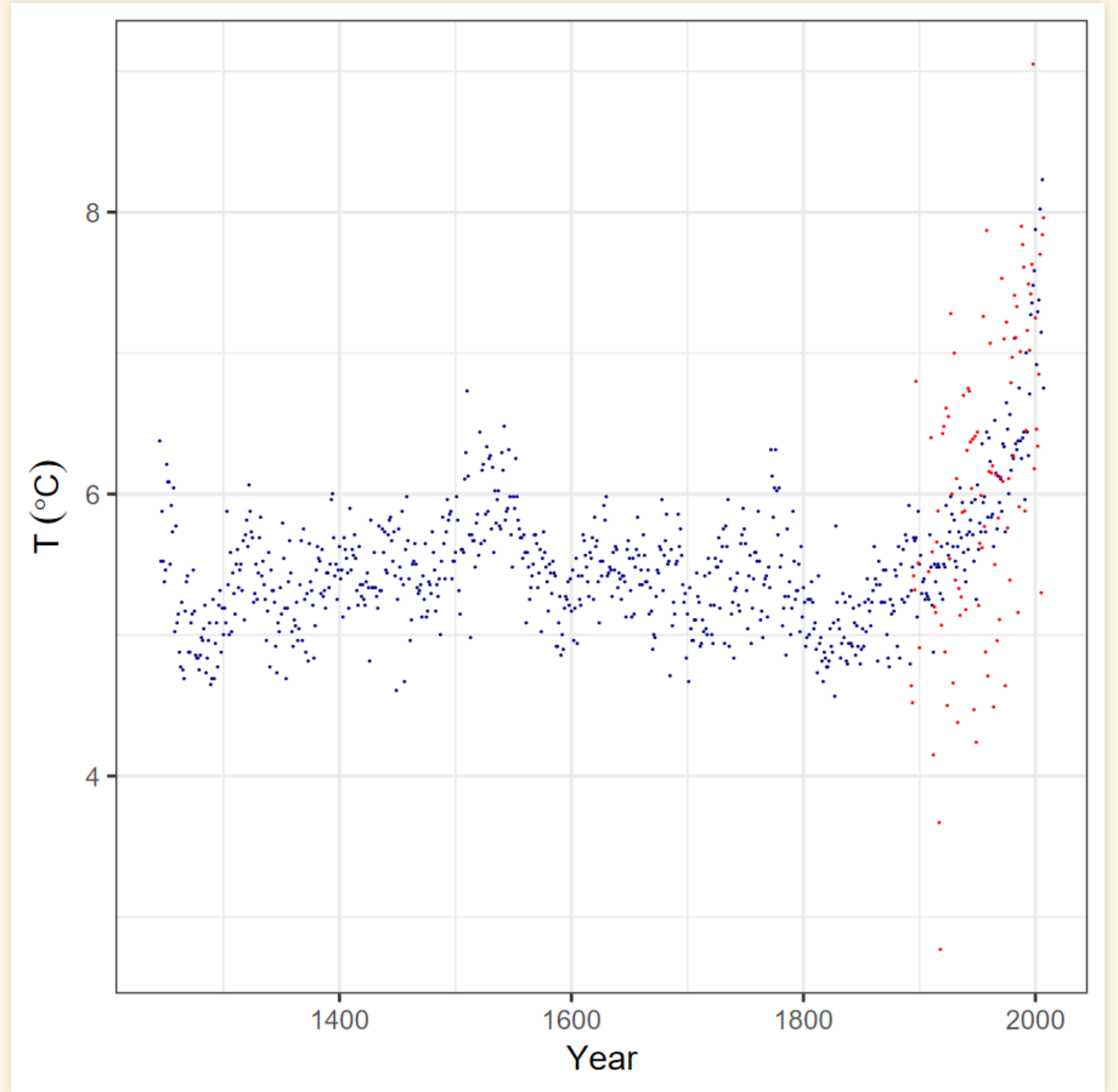
- Predictions

```
paleo <- paleo |> mutate(t_pred = predict(paleo_model, newdata =
paleo))
```

Plot Predictions

- Plot Predictions

```
ggplot(paleo, aes(x = year_ad, y = t_pred)) +  
  geom_point(size = 0.3, color = "darkblue") +  
  geom_point(aes(y = t_obs), size = 0.3, color = "red")  
  +  
  labs(x = "Year", y = expression(T~(degree * C)))
```



Assessing Model Quality

Regression Residuals

- Residuals are the differences between predictions $\hat{y}_i$ and observed y_i :

$$y_i = \hat{y}_i + \varepsilon_i$$

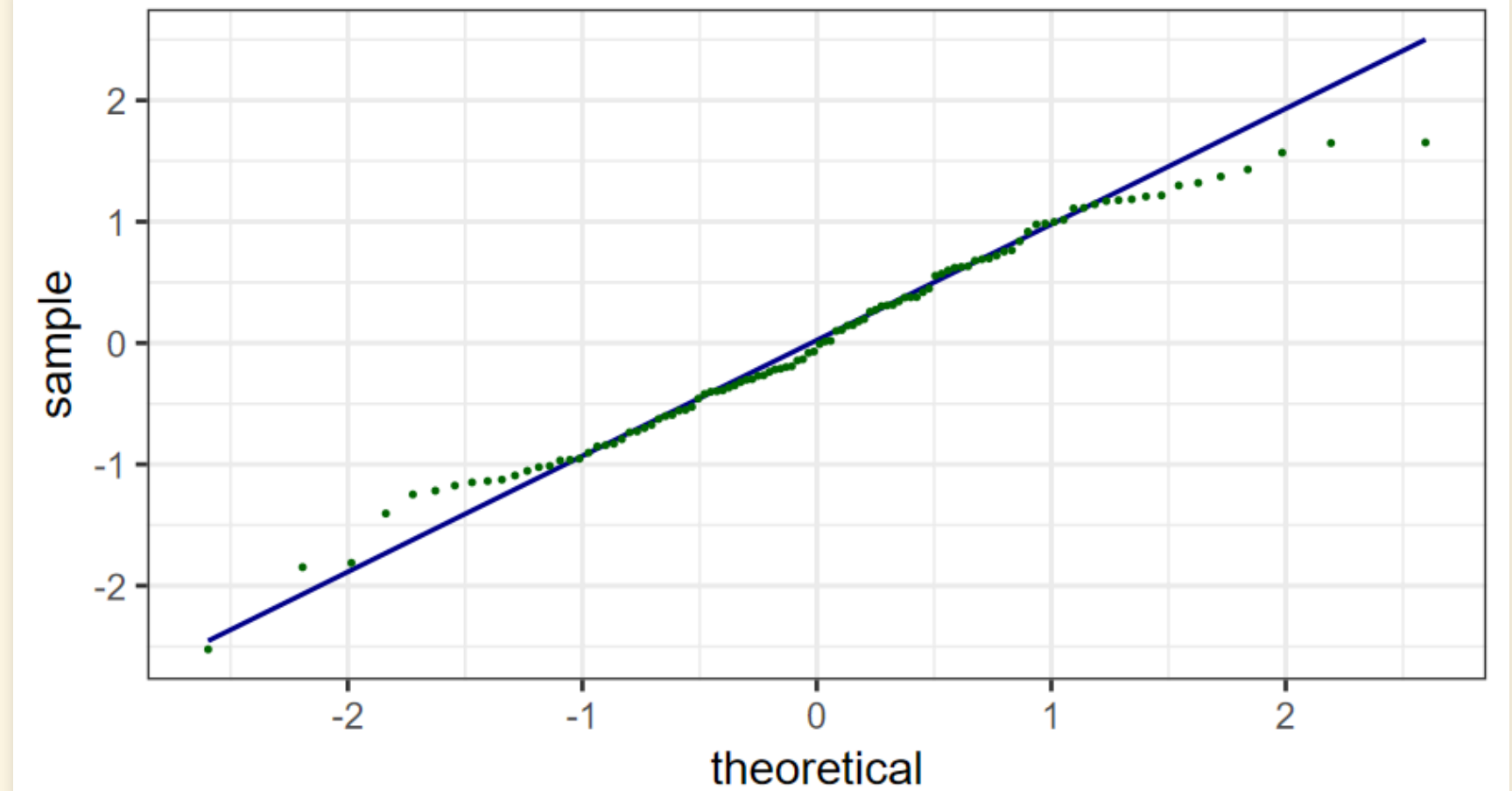
$$\hat{y}_i = \beta_0 + \beta_1 x_i$$

$$\varepsilon_i = y_i - \hat{y}_i$$

- Residual testing:
 - **Normality:** Residuals should be normally distributed (use Q-Q plot)
 - **Homogeneity:** Residuals should not show structure (e.g., _heteroskedasticity)
 - **Independence:** Residuals should be non-persistent (we'll cover this when we look at time series)

- Checking normality for tree ring data:

```
paleo_res <- paleo |> filter(! is.na(t_obs)) |>  
  mutate(res = t_obs - t_pred)
```

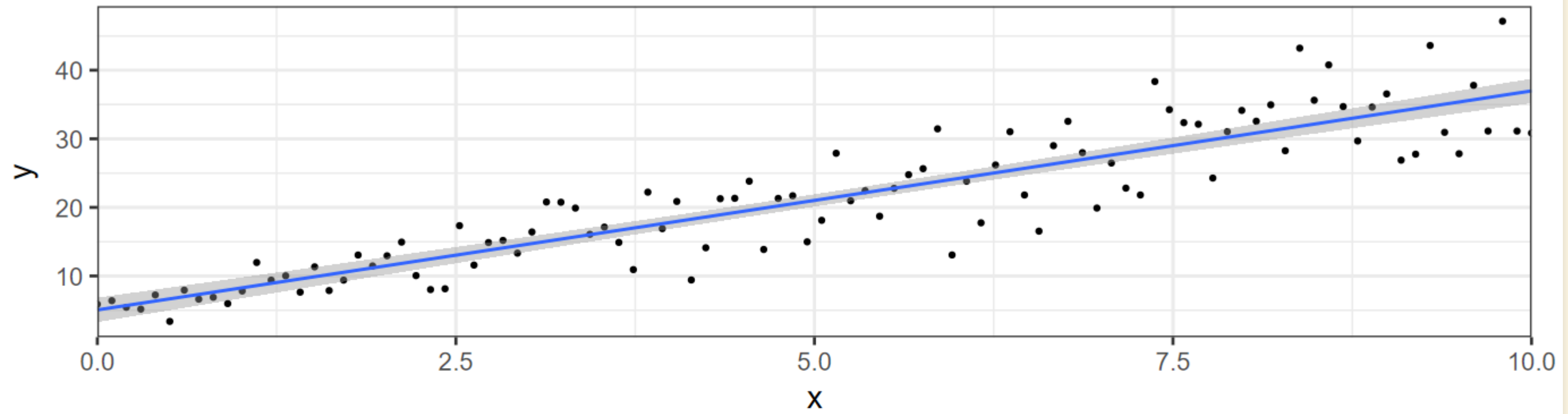


```
shapiro.test(paleo_res$res)
```

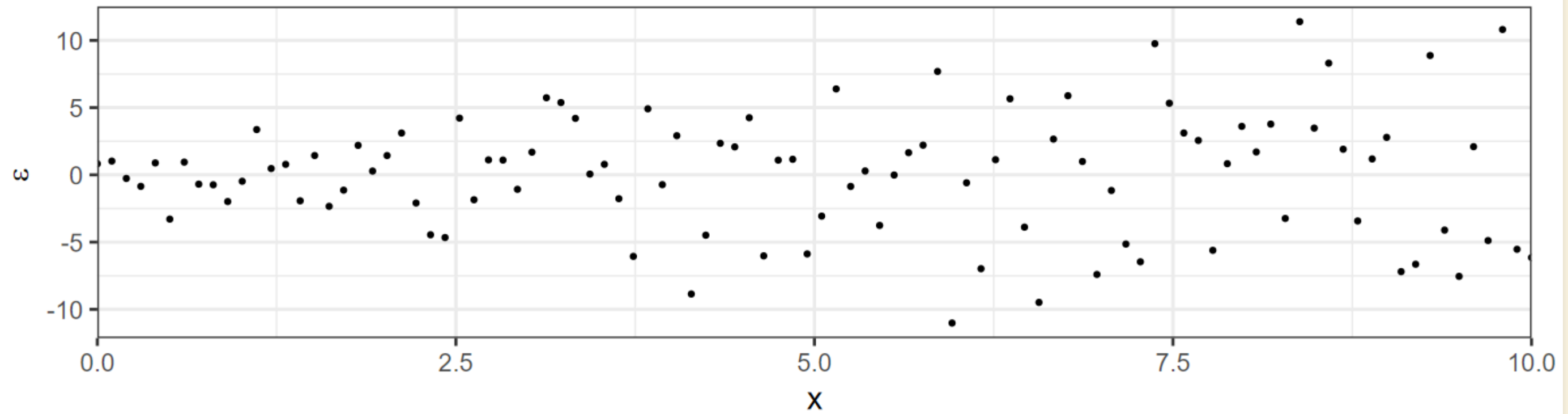
```
##  
##  Shapiro-Wilk normality test  
##  
## data:  paleo_res$res  
## W = 0.98369, p-value = 0.2206
```


Homogeneity

Heteroskedastic data

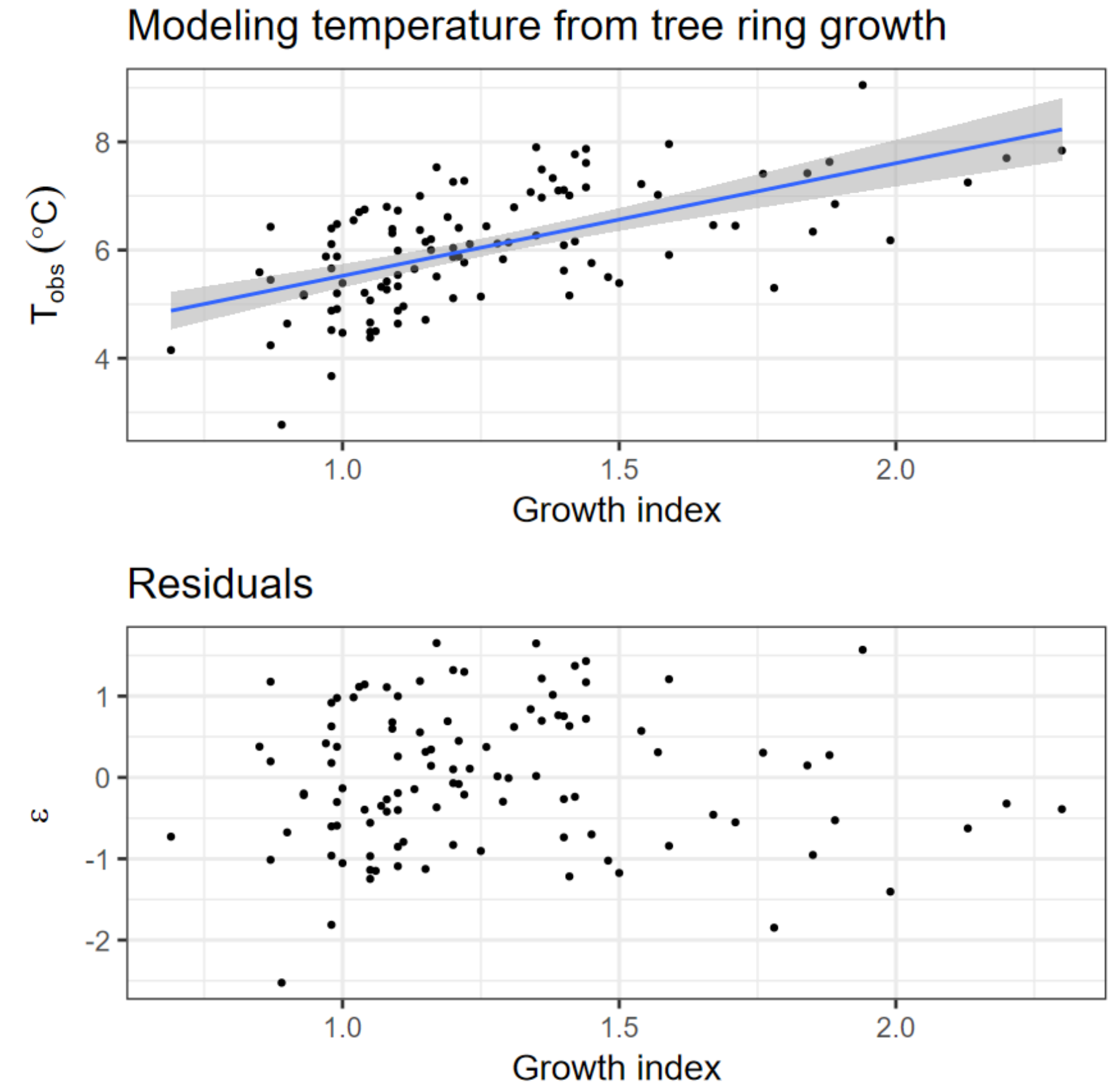


Residuals



Tree-Ring Data

```
p1 <- ggplot(paleo_res, aes(x = growth_index, y = t_obs))  
  +  
  geom_point() +  
  geom_smooth(method = "lm") +  
  labs(x = "Growth index", y = expression(T[obs]~(degree *  
    C)),  
    title = "Modeling temperature from tree ring  
    growth")  
  
p2 <- ggplot(paleo_res, aes(x = growth_index, y = res)) +  
  geom_point() +  
  labs(x = "Growth index", y = expression(epsilon),  
    title = "Residuals")  
  
p1 / p2
```



Analysis of Variance (ANOVA)

Analysis of Variance (ANOVA)

- Basic Model:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

$$\bar{y} = \frac{1}{N} \sum_{i=1}^N y_i \text{ (mean)}$$

$$\hat{y}_i = \beta_0 + \beta_1 x_i \text{ (prediction)}$$

$$\varepsilon_i = y_i - \hat{y}_i \text{ (error)}$$

- Mean Squared Error (MSE):

$$S_{\varepsilon}^2 = \frac{1}{N-2} \sum_{i=1}^N \varepsilon_i^2$$

- Sums of Squares:

$$\text{Total: SST} = \sum_{i=1}^N (y_i - \bar{y})^2$$

$$\text{Regression: SSR} = \sum_{i=1}^N (\hat{y}_i - \bar{y})^2$$

$$\text{Error: SSE} = \sum_{i=1}^N (y_i - \hat{y}_i)^2 = \sum_{i=1}^N \varepsilon_i^2$$

$$SSE = SST - SSR$$

$$S_{\varepsilon}^2 = \frac{1}{N-2} SSE = \frac{1}{N-2} (SST - SSR)$$

Using Analysis of Variance

- Summary:

$$SST = SSR + SSE$$

$$SST = (N - 1) \times \text{total variance in } y$$

$$SSR = (N - 1)\beta_1^2 \times \text{total variance in } x$$

$$SSE = \text{sum of squared residuals}$$

- ANOVA Table

Source	df	SS	MS	F
Total	$N - 1$	SST		
Regression	1	SSR	$MSR = \frac{SSR}{1}$	$\frac{MSR}{MSE}$
Residuals	$N - 2$	SSE	$MSE = \frac{SSE}{N - 2} = s_{\varepsilon}^2$	

- Mean squared error

$$MSE = s_{\varepsilon}^2 = \frac{1}{N - 2} SSE$$

- Coefficient of Determination R^2

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$$

R^2 measures the fraction of the total variance explained by the regression model

- F ratio:

$$F = \frac{MSR}{MSE}$$

Tree Ring Data

```
paleo_res |> summarize(
  SST = sum( (t_obs - mean(t_obs))^2 ),
  SSR = sum( (t_pred - mean(t_obs))^2 ),
  SSE = sum(res^2),
  R_squared = SSR / SST
)
```

```
## # A tibble: 1 × 4
##   SST    SSR    SSE R_squared
##   <dbl> <dbl> <dbl>   <dbl>
## 1  123.   44.6   78.3     0.363
```

- As expected, $SST = SSR + SSE$.
- R lets us generate an ANOVA table automatically:

```
anova(paleo_model)
```

```
## Analysis of Variance Table
##
## Response: t_obs
##           Df Sum Sq Mean Sq F value    Pr(>F)
## growth_index    1  44.562   44.562   59.197 8.517e-12 ***
## Residuals    104  78.289    0.753
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- What does this tell us?
 - $R^2 = 0.36$, so the regression model explains 36% of the total variance.
 - This is not a lot. There is a lot of variance in the air temperature that does not correlate with tree-ring growth.
 - But $p < 10^{-11}$, so we have a lot of confidence that the correlation between tree-ring growth and air temperature is real.
 - We can think of ε as “unexplained variance.”

Interpreting Unexplained Variance

Statistical and Systems Models

- Systems model: tree-ring growth depends on temperature, soil moisture, ...
- Statistical model

$$\text{growth} = f(T, M, \dots)$$

- The dependence is approximately linear near some reference values

$$\text{growth} = \alpha_0 + \alpha_T(T - T_0) + \alpha_M(M - M_0) + \dots$$

- We simplify and look only at the linear terms in T and ignore the rest.

$$\text{growth} = \alpha_0 + \alpha_T(T - T_0) + \dots$$

$$T = T_0 + \frac{1}{\alpha_T}(\text{growth} - \alpha_0 - \dots) \\ = \beta_0 + \beta_g \text{growth} + \varepsilon,$$

where ε represents all the terms in the real system that we're ignoring.

- Thus ε is the unexplained variance in T .
 - All the variance that isn't explained by the growth index.

